Areeb Noman

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EDUCATION

Maple Leaf International School — Primary School and High School (Science)
Grade: Playgroup – STD VII
Relevant Courses of High School: Physics, Chemistry, Biology, Mathematics B, Mathematics A, Pure Mathematics, English Language B, and Bangla
Leadership & Activities: Vice President, Science Club; Team Leader, Science Fair(3-time Champion)

Academia, Mirpur — High School (Science) and Secondary School (Science)
Grade: STD VIII – STD XII
Relevant Courses:
IGCSE: Physics, Chemistry, Biology, Mathematics B, Mathematics A, Pure Mathematics, English Language B, and Bangla
IAL: Physics, Chemistry, Pure Mathematics, and Further Mathematics

PROFESSIONAL EXPERIENCE

Kaz Software
June 2026 - Present
Software Engineering Intern

- Strengthened core programming fundamentals in Python (data structures, OOP, file handling, encryption) through structured guidance from senior engineers and mentors.
- Learned relational database concepts and SQL, gaining hands-on experience with MySQL/MariaDB — writing and debugging queries (INSERT, UPDATE, DELETE, JOIN), managing schema design, and handling data integrity issues (e.g., correct data typing, deduplication).
- Built and trained a deep learning pipeline for the RSNA Knee Abnormality Detection Kaggle competition to classify 12 clinical abnormalities from knee MRI studies. Designed and architected a ResNet50-based PyTorch model with multi-instance learning—extracting features from individual MRI slices and aggregating to study-level predictions. Owned all technical decisions including model structure, data preprocessing strategy (JPE image loading, grayscale-to-RGB conversion, medical image normalization), training optimization with checkpoint management and early stopping, and inference pipeline design. Implemented the complete end-to-end workflow with AI-assisted code generation.
- The Betting Scheme | Python, Tkinter GUI
  Designed the proportional payout algorithm and application structure, so that each bettor's share of the winning pool matches their share of the winning-side stakes.
  Developed and shipped the application using an AI-native workflow: I owned the design, logic, and decisions, and used AI to write and iterate on the code.

- ➢ Built a custom Android keyboard in Kotlin with three modes: Banglish to Bengali, Bengali to Banglish, and English. It uses rule-based transliteration and a frequency-ranked dictionary for real-time word suggestions. Developed AI-natively with Claude: defined requirements, tested on a physical Android device, and debugged layout, input, and romanization issues.
- ➢ Built an IP Locator tool in Python that accepts a user-provided IP address and retrieves its geographic data (city, country, latitude, longitude) via the ip-api.com REST API.
- ➢ Developed several supporting Python projects to reinforce fundamentals, including a password manager (symmetric encryption via the `cryptography` library), a terminal-based typing speed tester (`curses`), and text-processing tools.
- ➢ Built a personal portfolio to showcase project work and technical growth areebportfolio-tau.vercel.app

ACADEMIA
August 2024 - June 2026
Teaching Assistant (English)

- ➢ Directed 6+ English private tutoring batches, teaching 180+ IGCSE students of EDEXCEL.

EDIFY
July 2024 - June 2026
Assistant Teacher (Mathematics)

- ➢ Directed Mathematics private tutoring batches, teaching 300+ IGCSE and IAL students of both CIE and EDEXCEL.

SAT Math Tutor & Bootcamp Lead | Schoolhouse.world
January 2026 - Present
(World's largest free tutoring platform; in collaboration with College Board & Khan Academy)

- ➢ Launched and led SAT Math Bootcamps on the world's largest free global tutoring platform, delivering structured, exam-focused instruction.
- ➢ Guided students in Algebra, Advanced Math, and problem-solving strategies, improving SAT readiness, confidence, and performance.

Maple Leaf International School
Robotics Club, 2020
Technical Lead

- ➢ Conducted and supported robotics workshops and classes at school, engaging over 70+ students with an increase of 47% applications by the third workshop.

ACADEMIA
Science Club, 2022
Project Designer

- Designed interactive and engaging science projects to create awareness amongst the students about the cycles of Earth, for example, the water cycle, nitrogen cycle, etc. They helped students to understand science and the environment around them better, and grew their interest and curiosity about science. Created a project on energy-saving street lights, with a simple circuit excluding any need for codes, which was elected as the Best Project of the Year award.

ALOHA Bangladesh
2016 - 2019
Student

- Completed both Junior and Senior level courses of Aloha Mental Arithmetic with a grade of A+ and participated in multiple National level Mental Arithmetic Competitions hosted by ALOHA Bangladesh.

Josephite Math Mania
2019
Only qualifier from Maple Leaf International School

- Sole qualifier from my school. Participated in the National Round of both the Math Olympiad and the Sudoku Competition.

Banglar Math Team Championship
2024
Individual Participant and Team Lead

- Participated in the National Round for individual competition and led my team to qualify for both the Regional and National Round.

Scouts
2021 - 2022

- Cub Scout participant in school with a volunteering duration of 5 hours per week. Worked on campaigns for public road safety awareness, food and fundraising campaigns for the poor, volunteered for cleaning the roads of Dhanmondi, managed the annual sports day at my school, and worked in the cafeteria during school lunch break.

Prothom Surjo
2023 - Present

- Volunteered to educate helpless adolescent girls for the last 1.5 years who are extremely vulnerable due to being homeless and orphans, make them aware of the dangers wrapped within the society, including exploitation, and frequently being the victims of sexual

abuse and at risk of HIV/AIDS infection, physical torture, and trafficking, with a working period of 2 hours per week.

Volunteer for Bangladesh(VBD)
2024 - Present

- Volunteer work for Volunteer for Bangladesh (VBD), which is the youth wing of JAAGO Foundation, established in 2011. Volunteered for campaigns that include creating public awareness about the importance of poor hygiene, malnutrition, and a lack of basic health awareness and cleanliness, along with proper sanitation counselling under the surveillance of JAAGO Foundation medical authorities among the under privileged community; ensuring active participation of local people in dengue prevention, involving young people in environmental protection, distributing leaflets around dustbins and in open spaces, and creating awareness among the locals, especially during the periods of the Dengue epidemic.

STEM Leaders Program 2024, EMK Center
Participant & Appointed STEM Envoy (2024)

- I was selected to participate in the STEM Leaders Program 2024 at the EMK Center. I worked with other high school innovators, grades 9-12, in a three-day exploration of new technologies. Through hands-on workshops in virtual reality, 3D modeling, and programming, I improved my technical skills. I also took part in group challenges that focused on creativity, problem-solving, and community impact.

- After the program, I was named an EMK STEM Envoy. This is a six-month leadership role that encourages STEM education and innovation in local communities. This opportunity allows me to utilize my diverse strengths and mentoring experience to inspire others and promote change in STEM at the grassroots level.

Savvy Fellowship Program
2025 - Present

- Workshop about learning to gain product-market fit, acquire customers, and develop strategies for fundraising and scaling through visual presentations and interactive content.

IGAC Big Bang III 2025
Campus Ambassador and Team Leader(Fiscal Flash)

- Brought in participants for the event's segments. Participated in the Fiscal Flash round, solving the great Wadiyan Crisis, and ultimately led the team to the 3rd Runners Up position.

New York Academy of Sciences
Selected for Junior Academy for research project works and internships

International Computer Science Competition(ICSC)
Qualifier for the Semi-Final 2026

National High School Programming Contest(NHSPC)
National Round 2022
Qualifier for Python and C programming languages

ROSATOM Icebreaker Knowledge 2025
Ranked 4th in all of Bangladesh

Indonesian Mathematics Competition(IMC)
Merit enlisted

RESEARCH STUDIES

1. **BRUFAST(Biological Resource Usage with Fast Air Sucking Turbine):** Developed BRUFAST, a hybrid system that combines photosynthetic microorganisms (BRU) and compact air turbines (FAST) to tackle air pollution and energy shortages. The system is designed to be scalable, low-maintenance, and suitable for microgrids and public infrastructure. Proposed solutions focus on microbial health, energy storage, and mechanical durability to ensure long-term sustainability.

2. **BRUTE:** Developed a sustainable bioethanol production model using bacteria, taking advantage of their natural abilities and characteristics. Furthermore, it combines how bacteria can help with water treatment. Moreover, it encompasses the use of bacteria in medical nutrition, cosmetics, and the production of experimental biofuels. Showed skills in biotechnology, environmental engineering, and renewable energy systems.

INNOVATIVE PROJECTS

1. **Radiator cooling effect improvement:** Led project logistics and research support for a radiator cooling improvement initiative at the Gazipur 52 MW Power Plant. The goal was to boost engine efficiency in high ambient temperatures. I worked with engineering and operations teams to install a water-spray cooling system using the existing infrastructure. I also helped with performance analysis, documentation, and cost savings. The project resulted in reduced derating, improved thermal parameters, and increased power output, and was recognized as the **Project of the Year Award 2022,** securing BDT 300,000 (3 lakh) for implementation.

2. **Noise-Resilient Safety Alert System:** Designed and implemented an Arduino-based visual communication solution made from scratch, drafted for high-noise power-plant environments. The goal was to improve worker safety and communication where audio alerts were ineffective. I

developed a system using LED displays and rotating laser projection to transmit long-distance, non-audio alerts. I also contributed to testing, documentation, and deployment planning to ensure reliability under industrial conditions. The project reduced communication risks, enhanced safety protocols, and was recognized as the **Project of the Year Award 2023**, securing BDT 150,000 (1.5 lakh) for implementation.

HONOURS & AWARDS

- ➔ **Champion of the champions of BDMO 2021**
  Approximately the top 0.002% for that year. Won a prize money of 16,000 BDT from Prothom Alo.
- ➔ **The Daily Star Award 2024**
- ➔ **ACADEMIA High Achiever's Award 2024**

TRAINING & CERTIFICATIONS

1. **Harvard University – CS50's Introduction to Programming with Python**
   Completed an intensive programming course from Harvard, mastering Python fundamentals, problem-solving, and algorithm design.

SKILLS & EXTRACURRICULARS

Programming: C, Python
Soft Skills: Leadership, Public Speaking, Team Management, Critical Thinking
Interests: Scientific Research, Project Development, Community Development, STEM, AI, Computer Science
Languages: Bengali (Native), English (Fluent reading, speaking, and writing), Hindi(Speaking)